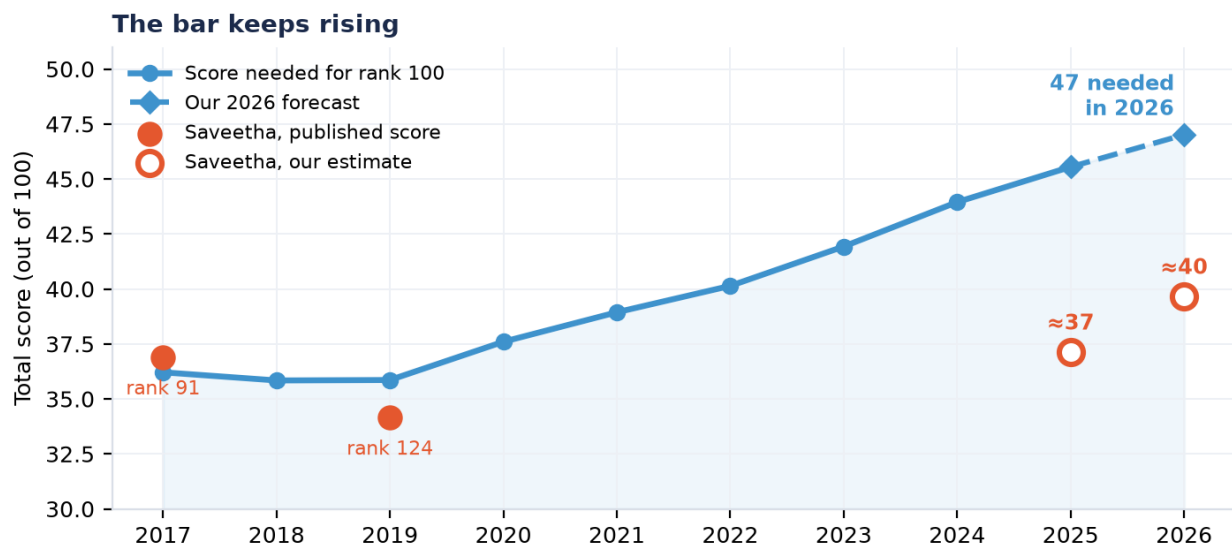
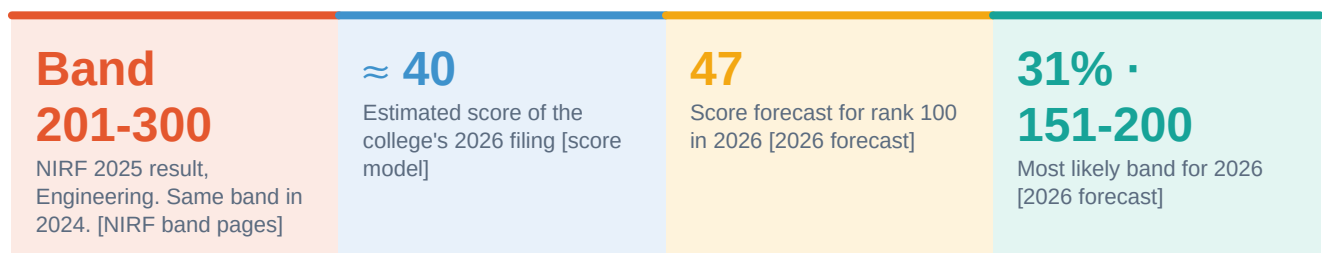


NIRF Analytics

Position, drivers and the 2026 outlook

An evidence-based reading of where the college stands in the NIRF Engineering rankings, how the score is built, and what the data says about 2026.

Prepared 16 September 2026 · Sources: nirfindia.org 2017-2025, NIRF methodology PDFs, the college's 2025 and 2026 filings
Every figure carries its source in grey. Grey boxes hold technical detail and can be skipped.



Blue line: the score needed to be ranked 100 each year. Red dots: the college's two published scores. Open circles: our estimates of its last two filings.

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025); Score model output, models/score_model.py → data/processed/model_report.json. Cut-off = lowest score in the top 100 of each year's Engineering table.

1 What the numbers say

- 1 Saveetha Engineering College is in the **201-300 band** of the NIRF Engineering ranking in both 2024 and 2025. NIRF gives no score for bands. [\[NIRF band pages\]](#)
- 2 The score needed for rank 100 rises every year: **41.93** in 2023, **43.95** in 2024, **45.55** in 2025. Our forecast for 2026 is about **47**. [\[NIRF ranking tables\]](#) [\[2026 forecast\]](#)
- 3 The college's own filings are worth about **37** (2025) and **40** (2026) by our model. The 2025 estimate falls inside the band NIRF actually published, a good sign the model reads the data correctly. [\[score model\]](#)
- 4 The college is **7 points** short of the 2026 rank-100 line. About **4 of those points** come from one parameter: Research (RPC). [\[analysis.json\]](#)
- 5 Behind the research gap: **12 PhDs a year** against 24 for institutes ranked 76-100, **61 full-time PhD scholars** against 163, and **Rs 30 lakh** of sponsored research a year against Rs 3.7 crore. [\[2026 filing\]](#) [\[institute PDFs\]](#)
- 6 Placements (97%), graduating on time (81%) and women students (33%) are already at or above the 76-100 tier. [\[2026 filing\]](#) [\[institute PDFs\]](#)
- 7 For 2026 the most likely result is the **151-200 band** (31%). Top 200: 58%. Top 100: 4%. [\[2026 forecast\]](#)

Every square bracket is a source. Section 10 lists where each one lives, with the exact web address or file.

2 2023, 2024 and 2025 at a glance

Year	Institutions that applied (Engineering)	Score at rank 100	Score at rank 1	Saveetha Engineering College	Saveetha Institute (the deemed university)
2023	1,314	41.93	89.79	Applied; not in the top 300	Rank 64 (49.30)
2024	1,463	43.95	89.46	Band 201-300	Rank 53 (52.85)
2025	1,584	45.55	88.72	Band 201-300	Rank 45 (56.55)

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025); nirfindia.org → Rankings → Engineering → Rank-band pages. 'Applied' counts come from the participant list on each year's page. The last column is a different institution, shown only to avoid confusion.

In 2023 the college applied but was not placed in any published list (NIRF published ranks 1-100 and bands to 200 that year). In 2024 and 2025 it was placed in the 201-300 band, which NIRF introduced in 2024 with 100 and 99 institutions respectively. Because bands carry no score, everything about the college's own numbers in this report is an estimate from its filings.

What changed between the two filings

What the college reported	2025 filing	2026 filing
Students (UG + PG)	5,196	5,961
Full-time faculty	390	405
Full-time PhD scholars	54	61
PhDs awarded per year (3-year average)	9	12
Graduating on time	61%	81%
Placed	92%	97%
Median UG salary	Rs 5.20 L	Rs 5.50 L
Operating spend per student	Rs 80,210	Rs 1.11 L
Sponsored research per year	Rs 25.24 L	Rs 29.83 L
Patents published (3 years)	173	460
Estimated total score (our model)	37.1	39.7

Source: College's NIRF 2025 Engineering filing, saveetha.ac.in/nirf-documents-2024 (Engineering-1.pdf); College's NIRF 2026 Engineering filing, saveetha.ac.in/SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf; Score model output, [models/score_model.py](#) → [data/processed/model_report.json](#).

3 How NIRF scores an engineering institution

Every institution gets a score out of 100. It is built from five parameters, each scored out of 100 and then weighted: [NIRF 2025 methodology, summary table]

$$\text{Total} = 0.30 \times \text{TLR} + 0.30 \times \text{RPC} + 0.20 \times \text{GO} + 0.10 \times \text{OI} + 0.10 \times \text{PR}$$

Parameter	Weight	What counts
TLR Teaching, Learning & Resources	0.30	How many students and faculty you have, how many faculty hold a PhD, and how much money you spend per student.
RPC Research & Professional Practice	0.30	Publications and citations per faculty, patents, and income from research projects and consultancy.
GO Graduation Outcomes	0.20	Share of students placed or in higher studies, share graduating on time, median salary, and PhDs awarded.

Parameter	Weight	What counts
OI Outreach & Inclusivity	0.10	Students from other states and countries, women students and faculty, fee-reimbursed students, facilities for the disabled.
PR Perception	0.10	A survey of employers and academics. Not based on submitted data.

Source: NIRF methodology PDF, Engineering 2025 (nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf). Weights from the 2025 Engineering summary table; 'what counts' paraphrases the sub-parameters, which are listed with marks and formulas in Appendix A.

WORTH KNOWING

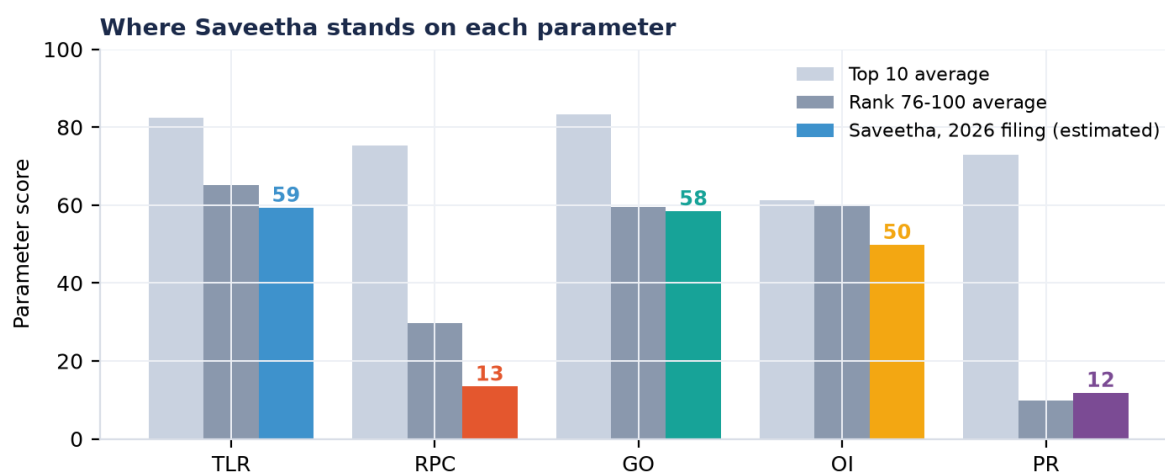
Two things that confuse people. Only the top 100 get a numeric rank; everyone below is placed in a band with no score. And **Saveetha Engineering College** (Sriperumbudur, NIRF id IR-E-C-16590) is a different NIRF entity from **Saveetha Institute of Medical and Technical Sciences** (Chennai, a deemed university, rank 45 in 2025).

4 The college's NIRF record, 2017 to 2025

Year	Category	Result	Total	TLR	RPC	GO	OI	PR
2017	Engineering	Rank 91	36.88	49.94	4.73	64.45	70.27	5.64
2018	Engineering	Band 101-150	—	—	—	—	—	—
2018	Overall	Band 151-200	—	—	—	—	—	—
2019	Engineering	Rank 124	34.14	55.68	2.99	55.82	53.45	0.41
2019	Overall	Band 151-200	—	—	—	—	—	—
2020	Engineering	Band 201-250	—	—	—	—	—	—
2021	Engineering	Band 201-250	—	—	—	—	—	—
2024	Engineering	Band 201-300	—	—	—	—	—	—
2025	Engineering	Band 201-300	—	—	—	—	—	—

Source: nirfindia.org → Rankings → Engineering, yearly tables (2017-2025); nirfindia.org → Rankings → Engineering → Rank-band pages. A dash means NIRF published no score. The college also applied in 2022 and 2023 without being placed in a published list.

The two years with published scores tell the same story as today: TLR, GO and OI were reasonable, **RPC was 4.7 and 3.0**, and Perception was close to zero.



Source: Score model output, `models/score_model.py` → `data/processed/model_report.json`; `analysis/analyze.py` → `data/processed/analysis.json` (computed from `db/nirf.db`). Coloured bars are model estimates from the 2026 filing; grey bars are averages of published 2025 scores.

5 The college next to the top 100, in plain numbers

	Top 10 (median)	Rank 76-100 (median)	Saveetha, 2026 filing
Students per faculty	14.7	14.6	14.9
Full-time PhD scholars	2,107	163	61
PhDs awarded per year	235.3	24	12
Graduating on time	81%	87%	81%
Placed	77%	70%	97%
Median UG salary	Rs 19.44 L	Rs 7.28 L	Rs 5.50 L
Operating spend per student	Rs 10.07 L	Rs 2.66 L	Rs 1.11 L
Sponsored research per year	Rs 154.87 Cr	Rs 3.66 Cr	Rs 29.83 L
Women students	20%	25%	33%
Students from other states	79%	57%	8%

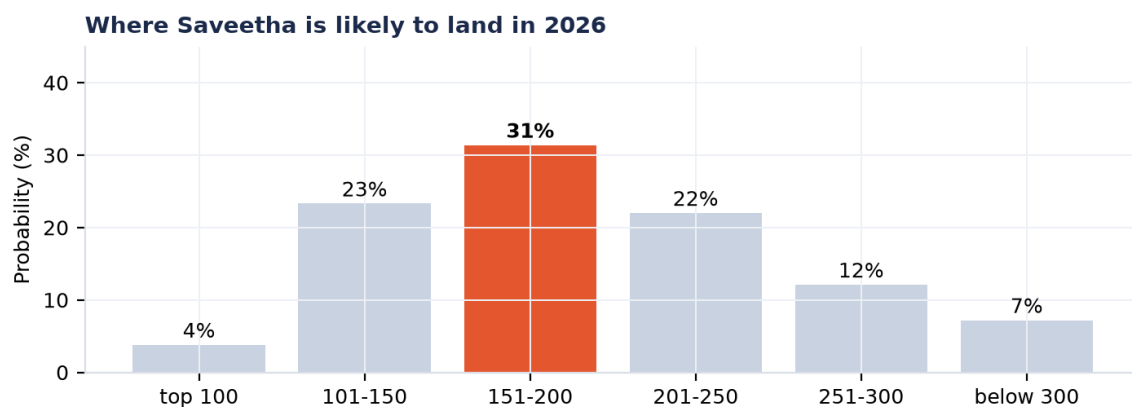
Source: nirfindia.org per-institute data PDFs (nirfpdcdn/pdf/Engineering/.pdf); College's NIRF 2026 Engineering filing, saveetha.ac.in (SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf); analysis/analyze.py → data/processed/analysis.json (computed from db/nirf.db). Medians over the 2025 top-100 institutes' own NIRF PDFs. Red = well below the 76-100 tier; green = at or above it.

The pattern is simple. Everything about research, PhDs and money per student is red. Everything about students, placements and inclusion is green. Put through the scoring weights, that is where the 7-point gap comes from: [analysis.json]

Parameter	Rank 90-100 average	Saveetha (estimate)	Points lost after weighting
TLR	64.5	59.3	+1.6
RPC	27.1	13.4	+4.1
GO	60.7	58.4	+0.5
OI	57.6	49.8	+0.8
PR	10.8	11.9	-0.1

Source: analysis/analyze.py → data/processed/analysis.json (computed from db/nirf.db); Score model output, models/score_model.py → data/processed/model_report.json. PR for the college is an assumption (median of private colleges ranked 60-100), not an estimate from data.

6 The 2026 outlook



Source: Forecast output, models/predict_2026.py → data/processed/prediction_2026.json. 20,000 simulations of the 2026 filing through the score model, with model error and an unknown Perception score, placed against the forecast thresholds.

40.12

Estimated total. Likely range
35.61 to 45.15

≈ rank 187

Expected position

58%

Chance of the top 200

6.9 pts

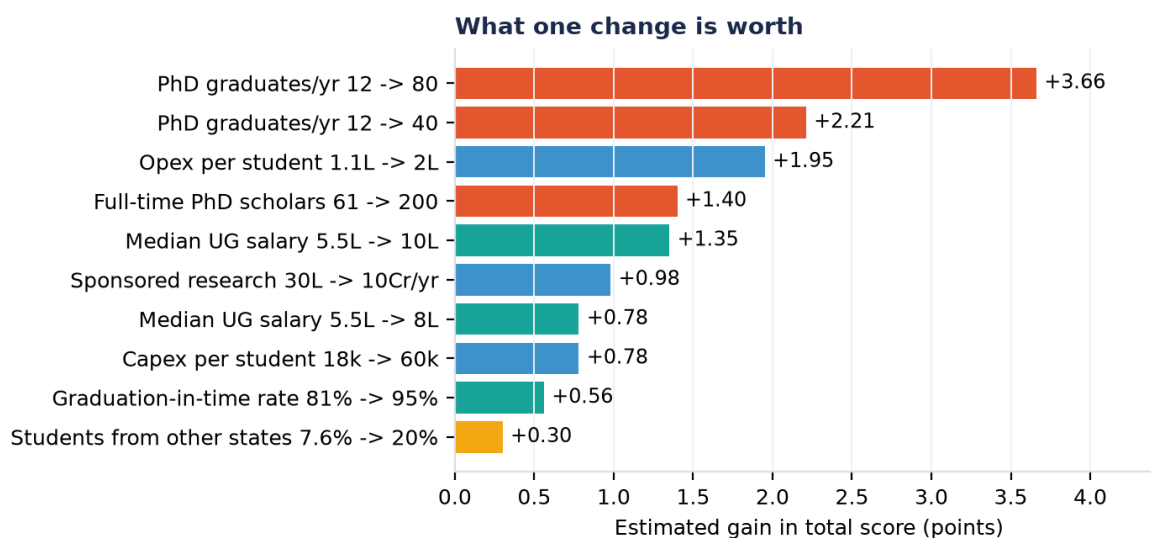
Gap to the top 100 (needs
about 47)

Rank in 2026	100	150	200	250	300
Score needed (forecast)	47.0	42.5	39.3	37.0	35.0

Source: Forecast output, models/predict_2026.py → data/processed/prediction_2026.json. Rank-100 line extended from the 2020-25 trend; the shape below rank 100 comes from 2019-22, when NIRF published 200 ranks.

The range is wide on purpose. Two things cannot be seen in any data: the Perception survey, and publication counts, which are 75 of Research's 100 marks. If the college's publication record is strong, the real result sits towards the top of the range.

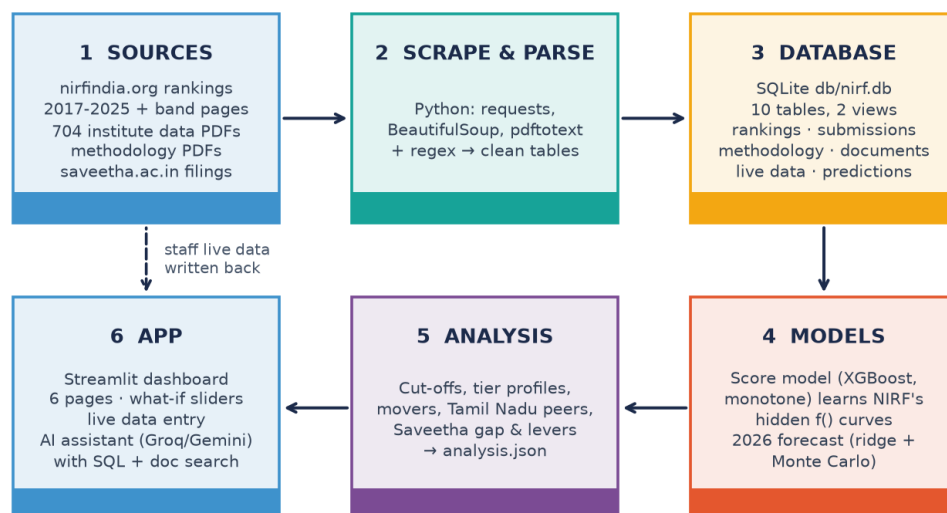
7 What moves the score



Source: Forecast output, models/predict_2026.py → data/processed/prediction_2026.json. Each bar is one change applied alone to the 2026 filing, re-scored by the model. Publications are not included because the PDFs do not contain them.

- **PhDs awarded** is the largest single lever: 12 → 40 a year adds about 2.2 points, 12 → 80 about 3.7. Awards lag enrolment by three to four years, so scholar intake now decides the 2029-30 filings.
- **Operating spend per student**: Rs 1.1 lakh → Rs 2 lakh adds about 2.0. NIRF counts labs, e-resources and equipment, not buildings.
- **Full-time PhD scholars**: 61 → 200 adds about 1.4.
- **Median salary**: Rs 5.5 lakh → Rs 8 lakh adds 0.8; → Rs 10 lakh adds 1.4. It is the median, so the middle of the batch matters more than a few high offers.
- **Sponsored research**: Rs 30 lakh → Rs 10 crore a year adds about 1.0.
- **Students from other states**: 7.6% → 20% adds about 0.3.
- **Publications and citations** are not in this chart because they are not in the data, yet they are 75 of RPC's 100 marks. The rank 76-100 tier averages RPC 30 against the college's estimated 13.

8 How this was done



One command (`./pipeline.sh`) re-runs every step when NIRF 2026 is published

Scripts download the public data from nirfindia.org and the college's website, turn the web tables and PDFs into clean rows, and store them in one database file. A model learns how NIRF turns raw numbers into scores from 700 institutes where both are known, then scores the college's own filings. A second model forecasts 2026. A web app shows all of it, lets staff type in live numbers, and has an AI assistant that answers questions from the data. One command re-runs everything when NIRF 2026 is published.

How accurate is the model?

Parameter	Institutes used	Share of variation explained (R^2)	Typical error
TLR	700	0.73	4.3 points
RPC	700	0.78	7.5 points
GO	700	0.82	3.7 points
OI	700	0.74	3.0 points

Source: Score model output, `models/score_model.py` → `data/processed/model_report.json`. Tested by hiding each institute and predicting it from the others. Combined with the published Perception score, the model reproduces published totals within 2.05 points on average.

KEEP IN MIND

The college's scores are **estimates**, not NIRF figures. Research is the least certain because publications are not in the PDFs. Perception is a survey and is treated as an assumption. The forecast assumes NIRF keeps the 2025 rules.

9 The app

Page	What it is for
Home	The band, the cut-off trend, the 2026 forecast, and how the top 100 earns its points
Top 100 Explorer	Every ranked Engineering institute 2017-2025, its parameter scores and the raw numbers it submitted
Saveetha Position	The college's history, estimated scores, and the gap against the tiers and Tamil Nadu peers
Gap & What-If	Sliders for every lever; the model re-scores instantly
Prediction 2026	Thresholds, band probabilities and model accuracy
Live Data Entry	Staff type in this year's numbers; they are kept with date and name, and the model re-scores with them
Ask the Data	An AI assistant that queries the database, reads the methodology, and can check nirfindia.org live. It shows every query it ran.

Run it with `./run.sh` and open `http://localhost:8501`. Refresh everything after NIRF 2026 with `./pipeline.sh`.

10 Sources and evidence

Every square bracket in this report points to one of these. Web addresses were fetched between 8 and 10 September 2026; copies of every page and PDF are kept unchanged under data/raw so any figure can be checked.

Tag	What it is	Where
NIRF ranking tables	Engineering (and Overall, College, University) tables 2017-2025 with rank, total and the five parameter scores	<a href="https://www.nirfindia.org/Rankings/<year>/EngineeringRanking.html">https://www.nirfindia.org/Rankings/<year>/EngineeringRanking.html
NIRF band pages	Institutes placed in 101-150, 151-200, 201-300 (no scores)	<a href="https://www.nirfindia.org/Rankings/<year>/EngineeringRanking150.html">https://www.nirfindia.org/Rankings/<year>/EngineeringRanking150.html , ...200.html, ...300.html
institute PDFs	The data each ranked institute submitted (704 PDFs: top 100 for 2023-25, top 200 for 2021-22)	<a href="https://www.nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<institute id>.pdf">https://www.nirfindia.org/nirfpdfcdn/<year>/pdf/Engineering/<institute id>.pdf
NIRF 2025 methodology	Official parameters, marks, weights and formulas (also 2023 and 2024 versions)	https://www.nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf
2025 filing	The college's NIRF 2025 Engineering submission	https://saveetha.ac.in/nirf-documents-2024/ → Engineering-1.pdf
2026 filing	The college's NIRF 2026 Engineering submission (uploaded February 2026)	https://saveetha.ac.in → NIRF → SAVEETHA-ENGINEERING-COLLEGE20260216-.pdf
score model	Estimated parameter scores and accuracy	models/score_model.py → data/processed/model_report.json
2026 forecast	Thresholds, band probabilities, levers	models/predict_2026.py → data/processed/prediction_2026.json
analysis.json	Cut-offs, tier profiles, medians, gap table	analysis/analyze.py → data/processed/analysis.json
database	All of the above in one file	db/nirf.db (tables: rankings, rank_bands, participants, submissions, methodology, documents, faculty, predictions, saveetha_live)

A Appendix: sub-parameters and formulas

Parameter	Sub-parameter	Marks	In plain words
TLR (weight 0.30)	Student Strength including Doctoral Students (SS)	20	Number of students, incl. PhD scholars
TLR (weight 0.30)	Faculty-student ratio with emphasis on permanent faculty (FSR)	30	Faculty-student ratio; 1:15 earns full marks
TLR (weight 0.30)	Combined metric for Faculty with PhD (or equivalent) and Experience (FQE)	20	Share of faculty with a PhD, and the experience mix
TLR (weight 0.30)	Financial Resources and their Utilisation (FRU)	30	Money spent per student, capital and operating
RPC (weight 0.30)	Combined metric for Publications (PU)	35	Publications per faculty (Scopus / Web of Science)
RPC (weight 0.30)	Combined metric for Quality of Publications (QP)	40	Citations and top-25% papers per faculty
RPC (weight 0.30)	IPR and Patents: Published and Granted (IPR)	15	Patents published and granted
RPC (weight 0.30)	Footprint of Projects and Professional Practice (FPPP)	10	Research funding and consultancy income
GO (weight 0.20)	Combined metric for Placement and Higher Studies (GPH)	40	Share placed plus share going to higher studies

Parameter	Sub-parameter	Marks	In plain words
GO (weight 0.20)	Metric for University Examinations (GUE)	15	Share graduating on time; 80% earns full marks
GO (weight 0.20)	Median Salary (GMS)	25	Median salary of placed graduates
GO (weight 0.20)	Metric for Number of Ph.D. Students Graduated (GPHD)	20	PhDs awarded per year
OI (weight 0.10)	Economically and Socially Challenged Students (ESCS)	20	Students on full fee reimbursement
OI (weight 0.10)	Facilities for Physically Challenged Students (PCS)	20	Facilities for physically challenged students
PR (weight 0.10)	Peer Perception: Employers & Academic Peer (PR)	100	A survey of employers and academics

Source: NIRF methodology PDF, Engineering 2025 (nirfindia.org/nirfpdfcdn/2025/framework/Engineering.pdf). Marks as printed in the 2025 Engineering methodology.

TECHNICAL NOTE — SKIP IF YOU LIKE

Formulas as printed in the 2025 Engineering methodology. $FSR = 30 \times [15 \times (F/N)]$, F = full-time faculty, N = students incl. PhD scholars; zero below 1:50. $FQE = 10 \times (FRA/95)$ for the PhD share plus up to 10 for an even spread over ≤ 8 , 8-15 and >15 years of experience. $FRU = 7.5 \times f(BC) + 22.5 \times f(BO)$ on capital and operating spend per student. $PU = 35 \times f(P/FRQ) - 5 \times f(Pret)$; $QP = 20 \times f(CC/FRQ) + 20 \times f(TOP25P/P) - 5 \times f(Cret)$; the -5 terms for retracted work are new in 2025. $IPR = 10 \times f(patents\ granted) + 5 \times f(patents\ published)$. $GPH = 40 \times (Np/100 + Nhs/100)$. $GUE = 15 \times \min(Ng/80, 1)$. $GMS = 25 \times f(\text{median salary})$. $GPHD = 20 \times f(\text{PhDs graduated})$. $RD = 25 \times (\text{share from other states}) + 5 \times (\text{share from abroad})$. $WD = 15 \times (\text{women students \%}/50) + 15 \times (\text{women faculty \%}/20)$. $ESCS = 20 \times f(\text{fee-reimbursed \%})$. The $f()$ curves are NIRF's own normalisation and are not published; the score model learns them from data.

TECHNICAL NOTE — SKIP IF YOU LIKE

Model detail. One gradient-boosting model (XGBoost) per parameter, trained on 700 institute-years (ranks 1-200, 2021-25) with monotone constraints so that a better input can never lower a score. Inputs per parameter follow the methodology (TLR: students, PhD scholars, faculty per student, spend per student; RPC: PhDs awarded, sponsored and consultancy income, faculty size; GO: placed-or-higher-studies share, on-time graduation, median salary, PhDs awarded; OI: out-of-state share, women share, fee-reimbursed share, disability facilities). Grouped 5-fold cross-validation. 2026 forecast: linear trend of the rank-100 cut-off since 2020 (+1.616 per year), score-vs-rank shape from 2019-22, 20,000 Monte-Carlo draws over model error and Perception; a ridge model on previous-year scores projects the rest of the field (validated on 2025: error 1.36 points).

Glossary

Term	Meaning
NIRF	National Institutional Ranking Framework, the Ministry of Education's ranking
Band	A range such as 201-300 used instead of a rank for institutes below 100; no score is published
Cut-off	The lowest score inside the top 100 in a given year
Filing	The data form an institution uploads to NIRF, published as a PDF
Estimate	A score computed by our model, not by NIRF
R ²	Share of the variation the model explains; 1.0 is perfect